

Proposal review form

April 18, 2026

General information

Title of the proposal

Research proposal: Testing a spectral epidemic threshold for SIS on networks under heavy-tailed recovery

Authors of the proposal

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1 Part 1

The questions in this part probably take several few lines to answer. Make sure to explain your answer.

1.1 Give two positive general points about the proposal

First, the proposal identifies a concrete gap between widely used spectral mean-field thresholds ($\tau_c \approx c/\lambda_{\max}$) and more realistic recovery-time variability, and it motivates why that matters for interpreting whether an infection is “under control” on a given contact structure. The link to recent work on generalised recovery (including heavy tails) makes the problem feel timely rather than purely technical.

Second, the empirical strategy is well aligned with the questions: a baseline under exponential recovery (**RQ1**), controlled shifts to heavy-tailed recovery at fixed mean infectious period (**RQ2**), and an explicit plan to separate threshold shifts from extinction-time and variability effects (**RQ3**). The risk analysis also shows awareness of discrete-time versus continuous-time theory and of stochasticity near the threshold.

1.2 Give two general points to be improved in the proposal

One limitation is generalisability: the design fixes a single Erdős–Rényi draw and attributes differences mainly to the recovery distribution. That is clean for internal validity, but the reader may wonder how sensitive conclusions are to graph realisation, size, and mean degree. A short note on whether (or how) you might replicate across independent network samples would strengthen the design story without necessarily expanding scope much.

Second, the operational definition of the “empirical threshold” could be spelled out more precisely in the proposal itself—for example, how “typically persists” is turned into a rule (survival fraction at a horizon, quasi-stationary estimates, grid resolution, number of replicates). Right now the intent is clear, but a reviewer will look for enough detail to judge feasibility and reproducibility of that estimation step.

1.3 Is it generally clear what this proposal wants to investigate? Why (not) ?

Yes. The central aim is to test, in an agent-based SIS simulation on a random network, whether a simple spectral mean-field threshold remains a good benchmark when recovery times are heavy-tailed rather than exponential, holding the mean recovery time fixed—and to see whether effects show up mainly in the threshold location or in persistence variability and extinction times. The background section explains the standard picture and why heavy tails may break it, so the investigation is easy to follow.

1.4 Are the research questions and hypotheses clear and properly formulated? Are the concepts used in them operationalised in good fashion?

The hypothesis (heavy-tailed recovery facilitates persistence relative to the standard rule of thumb at fixed mean recovery time) is stated plainly. **RQ1–RQ3** form a logical sequence from validating the benchmark case to comparing distributions to disentangling threshold shifts from dynamic variability. Key quantities—effective transmission τ , spectral reference $\tau_c \approx 1/\lambda_{\max}$, infection probability sweeps, and logged outcomes (extinction vs. late-window prevalence)—are tied to the questions in the methods section. As noted above, tightening the threshold-estimation procedure in one short paragraph would complete the operationalisation for an external reader.

1.5 What is your impression of the research setup/design? Can the research questions be answered this way?

The setup is appropriate: NetLogo gives an explicit individual-based SIS process on a network; Python supplies $\lambda_{\max}$ for the same exported graph, which

keeps the comparison between “prediction” and “simulation world” consistent. Sweeping infection strength while fixing the graph and mean recovery time is the right control structure for **RQ2**. **RQ3** is answerable if the analysis plan contrasts not only point estimates of thresholds but also distributions of extinction times and prevalence variation—as the proposal already suggests. The main caveat is stochasticity and finite-time effects near criticality, which you partly mitigate via repetitions; the feasibility of obtaining stable threshold estimates within the time budget should be monitored.

1.6 Is the domain/simulation chosen relevant and/or interesting and is it clearly described?

Yes. Network SIS models with non-exponential recovery are an active modelling line, and the spectral threshold is a standard reference in network epidemiology, so the domain is relevant. The simulation domain (Erdős–Rényi contact network, SIS dynamics, three recovery families) is described clearly enough to understand what is being compared and what is held fixed.

1.7 Give some suggestions to the authors on how to improve the final report if they would recycle the text from the proposal

When expanding to the full report, add a concise “Estimation protocol” subsection: explicit survival/persistence criterion, replication counts, grid refinement (coarse-to-fine), and how uncertainty will be shown (intervals across runs or networks). If space allows, one supplementary figure or table on robustness across a few independent ER realisations (same N and edge probability) would greatly help readers trust that findings are not an artefact of one graph draw. Finally, make the mapping from discrete-time per-tick probabilities to the continuous-time τ notation explicit, so the comparison to theory is transparent.

2 Part 2

The questions in this part probably require a single line to provide feedback to. Try to say a bit more than just yes or no (or partially). In other words, explain your answers.

2.1 What is the quality of the language/style/spelling used?

The language is fluent, academic, and readable. Terminology from epidemiology and network science is used consistently, and the narrative flows from motivation to questions to methods without abrupt jumps.

2.2 What is the quality of the used terminology?

Strong. Concepts such as SIS, endemic threshold, adjacency spectrum, heavy-tailed recovery, and effective transmission are introduced at an appropriate level and used coherently with the cited literature.

2.3 Did the authors make good use of literature?

Yes. The proposal anchors the spectral threshold story in recognised network-epidemiology references and connects to newer work on generalised recovery-time structure, which supports both the novelty claim and the modelling choices.

2.4 Is the proposal written with attention to detail?

Mostly yes: parameters to vary and hold fixed are listed, outputs per run are specified, and risks (discrete time, workload, stochasticity) are discussed. The remaining detail gap is the precise algorithmic definition of the empirical threshold and the planned sample sizes, which are implied more than fully specified.

2.5 Is the proposed research setup clear enough?

Clear at the level of tools (NetLogo, Python), network type, disease process, and comparison logic. A reader could still ask for pseudocode-level detail on the heavy-tailed recovery implementation and the sweep/replication schedule; adding that in the final report would help reproducibility.

2.6 Is it fun/interesting to read?

Yes for an audience interested in epidemic modelling and networks: the question is well motivated, the tension between simple rules and realistic recovery variability is intuitive, and the proposed simulations are a natural way to probe that tension.

2.7 Is the planning of the proposal adequate?

The week-by-week outline matches the staged execution of **RQ1–RQ3** and leaves time for analysis and writing. It is somewhat compressed on paper; success will depend on automation, early pilot runs, and possibly narrowing sweep ranges after coarse exploration—which you already hint at under risks.

2.8 Is the proposed data analysis feasible?

Feasible in principle: merging BehaviourSpace-style outputs and plotting prevalence curves, threshold comparisons, and variability summaries is standard. The heavy lifting is computational (many runs) and statistical (stable estimation near criticality), not exotic analysis. The main feasibility risk is runtime and data volume, not the complexity of the analysis code itself.